

05 Host a Web Site with TLS

Goals

- Exercise your Linux command line skills
- Learn how to add a certificate to a web server
- Understand TLS in practice

Steps

1. Get a VM from vcm.duke.edu. We've already talked about this.
2. Get a web server. You can use what you want, but I made it work with Apache, which you can get with `sudo apt-get install -y apache2`.
3. Run web server. If you took the last hint, Apache is already running. You can check on it with `sudo service apache2 status`.
4. Get certificate and private key. You can use whatever you like, but you shouldn't have to pay for it. I made it work with <https://letsencrypt.org/> using their Certbot program.
5. Install certificate and key in web server. Certbot actually took care of everything for me so I didn't even have to explicitly do this step.
6. Verify TLS functionality

Submission

Your submission should include the following items:

1. Screen shot of your browser having a successful HTTPS connection to your web server
2. Your web server certificate chain as a file

In addition, please answer the following questions.

1. What web server and version did you run?
2. What TLS implementation and version does your web server use? How did you find this information?
3. (Extra credit) What TLS version and cipher suite is used when you connect from your browser? How did you find out?
4. What TLS version and cipher suite do you get when you connect using `openssl s_client` at the command line?
5. Force a different TLS version using `openssl`. Show your command and the output.
6. Force a different cipher suite using `openssl`. Show your command and the output.
7. (Optional) If you are able to leave your web server running, give me the URL so I can try it out myself